# Who Needs a Savings Nudge? Leakage-Aware Prediction of Household Savings Adequacy in India

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***Abstract*—A savings-product or financial-inclusion team cannot review tens of thousands of households by hand, so triage has to be automated. We study it on the India Human Development Survey-II (41,518 households, 2011–12), predicting whether a household retains at least 20% of its annual income after recorded consumption. The label is an accounting identity, so expense-to-income ratios reconstruct it at 99.75% agreement. Spending shares look safer but also leak: food spend tracks household size, so a formula from household size, food share and income ranks households at ROC-AUC 0.895 with no fitted model. A simulation shows the leak depends on one regression residual, and HCES 2022–23 shows the same mechanism. The headline model therefore uses income and demographics only: ROC-AUC 0.882 under cross-validation grouped by sampling unit and 0.886 on held-out units. At a 25% contact budget it reaches recall 36.2% against a ceiling of 36.7% (precision 98.5%), 1.1 points above poorest-first (95% CI 1.0–1.3). Whether that helps depends on who responds to a nudge, which the survey cannot identify. In an explicit cost-and-response what-if, a risk ranking moves fewer households than random contact, because the most at-risk are out of reach, while a reachability ranking moves more. Rankings transfer across states and sectors; calibration does not.**

***Index Terms*—Machine learning, financial inclusion, household survey data, data leakage, compositional data, explainable AI, grouped cross-validation**

## I. Introduction

Savings products in India are sold into a population that is mostly not saving. Outreach is expensive per contact, budgets are fixed, and a team that wants to prioritise households by need has no way to inspect tens of thousands of them individually. The decision is a triage: treat this household as on track or at risk, and spend the next contact accordingly.

We frame that triage as binary classification on the India Human Development Survey-II (IHDS-II), a nationally representative survey of 42,152 Indian households conducted in 2011–12 [1]. IHDS-II is unusual among Indian surveys in recording income and category-level expenditure for the same household; the NSS-lineage consumption surveys collect one without the other. After dropping 634 households with non-positive income or missing consumption, 41,518 remain, in 2,461 primary sampling units (PSUs).

The problem is constrained in a way that is easy to miss. A household is labelled on track when it retains at least 20% of its annual income, and that label is an arithmetic function of income and total consumption. Expense-to-income ratios, the intuitive representation, reconstruct the answer at 99.75% agreement. Composition shares describe only the shape of a budget and pass every single-feature check, but jointly they are not safe: given household size and income, the food share approximately recovers the size of the budget, and with it the label. The paper is organised around that finding and around a second constraint on reporting: against the rule a team would use without a model, contact the poorest first, most of the apparent gain disappears.

### A. Scope and objectives

The project has five objectives: establish by measurement, univariate and multivariate, which representations of household spending can legitimately serve as features for a savings-adequacy target, and when they leak; compare seven model families under PSU-grouped validation fixed in advance and quantify the gain over a single-variable rule rather than over chance, reporting recall, precision and ceiling capture separately; explain the winning model with exact tree SHAP values grouped into feature families; test how the targeting decision depends on cost and on who responds to a nudge, without claiming the survey identifies response; and test how far the findings travel beyond 2011–12 and the surveyed states.

## II. Related Work

Predicting household welfare from survey covariates is the core of proxy means testing, the standard method for targeting transfers when income is unobservable. Brown, Ravallion and van de Walle evaluated such tests across nine African countries and found that they miss a large share of the poorest households even when the underlying regression fits well [2], which is the closest antecedent to our finding that a good score does not imply a useful targeting instrument. McBride and Nichols showed that out-of-sample validation and machine learning improve on the regression practice then in use [3], and where survey data is unavailable poverty has been predicted

from mobile phone metadata [4]. Our task differs in what is predicted: the outcome is a savings residual rather than a consumption level, which is what makes spending-based features dangerous.

Ours is a prediction policy problem in the sense of Kleinberg et al. [5]: we do not claim that contacting a household changes its savings rate, only that ranking by predicted risk allocates a fixed budget better than not ranking. The intervention literature supplies the reason such a ranking is worth having, since simple reminders raise savings measurably [6], while Dupas and Robinson show that access constraints rather than preferences explain much of the shortfall among poor households [7].

Two measurement results bear on the data. Deaton and Kozel document the divergence between income and consumption aggregates in Indian surveys [8], which we inherit as 55.9% of households reporting consumption above income. Engel's law, food's budget share falling as income rises, appears here at Spearman $\rho = -0.280$.

Leakage is a known failure of data-mining studies [9], usually traced to a single column; here it arises only jointly. On method we use gradient-boosted trees [10] with SHAP attributions from the exact tree algorithm [11], [12]. Because the eleven spending shares are a composition, we follow Aitchison's treatment of the simplex [13], cluster on an isometric log-ratio basis [14] and replace zeros multiplicatively [15]; cluster count is chosen by silhouette [16], Davies–Bouldin and Calinski–Harabasz indices.

## III. Methodology

### A. Data and target

The analysis file is built from ICPSR study 36151, dataset DS0002 (Household). IHDS-II records consumption in 52 items under two recall windows, 30-day and annual, so the build annualises as $12 \times$ monthly $+$ annual and maps items onto 11 expense categories from the codebook. The 11-category total matches the survey's own aggregate `COTOTAL` with correlation 0.9932, median difference zero and 97.7% of households within 1%; a label computed from it agrees with ours for 99.75%. Occupation is the worker type contributing the most workers, ties (18% of households) going to the more regular income stream. A missing outstanding-debt value is kept as missing, with an indicator, rather than read as zero.

The target is

$$\texttt{Goal_Met} = \mathbf{1}\left[\frac{\texttt{INCOME} - \texttt{COTOTAL}}{\texttt{INCOME}} \geq 0.20\right], \qquad (1)$$

with both quantities in annual rupees. The rate is 31.93% positive, a 2.13:1 imbalance, and 30.47% when survey weights are applied. The 20% figure is a convention, so we publish the sensitivity curve: the at-risk share moves from 68.1% at the benchmark to 61.7% at 10% and 74.7% at 30%, with about 93.5% label agreement either way. Because IHDS records no savings intention, the benchmark is normative and identical for every household; the model measures savings capacity, not goal alignment.

### B. The leakage check

Equation (1) is exact over `INCOME` and `COTOTAL`, and approximate over the 11 categories. The raw rupee categories, with income, reconstruct the label at 99.75% agreement, and expense-to-income ratios give the same 99.75%. Both are excluded.

Composition shares sum to one, so no single share carries the label: the largest marginal ROC-AUC distance from 0.5 is 0.100 (healthcare share). The multivariate route is different. Food spending rises with household size (elasticity 0.63, $R^2 = 0.29$ in logs), so $\texttt{COTOTAL} \approx k \cdot \text{size}/\text{food share}$, and the score $1-(k\cdot\text{size}/\text{food share})/\texttt{INCOME}$, with $k$ the median food spend per person (Rs. 8,848 a year), ranks households at ROC-AUC 0.895 with no fitted model. It excludes only the 24 households (0.06%) with a zero food share. In an ablation under grouped cross-validation, dropping the food share alone costs 0.002 AUC (the other shares substitute for it), dropping the household-size family costs 0.010, and dropping all shares and participation indicators keeps 94.3% of the full model's AUC (0.878 against 0.931). We fixed a decision rule before running these checks: if the formula scored above 0.85, or a model without shares kept most of the full model's AUC, the headline model would exclude spending shares. Both conditions hold.

Six survey-reported indicators of savings-instrument holding sit outside the consumption arithmetic and are reserved as an external check: fixed-deposit holding is 15.2% among on-track households against 8.2% among the rest.

### C. Features and validation

Two feature sets are carried. The *deployable* set (13 features) is log income, household size, age dependents (members aged 0–14 or 60+), dependency ratio, head age, highest adult education, debt-to-income, a debt indicator, a missing-debt indicator and four one-hot categoricals (occupation, area type, caste, religion) with an explicit `Unknown` level. The *full* set (29) adds the 11 shares and five participation indicators for majority-zero categories.

All design decisions were taken on a training split drawn first: one fold of `StratifiedGroupKFold` grouped by PSU, leaving 8,299 test households in 472 PSUs that no decision saw. Everything else uses five-fold stratified cross-validation grouped by PSU on the remaining 33,219 households, because neighbours share prices, labour markets and interviewers. Debt is winsorised at the 99th percentile inside the pipeline, refit per fold. Linear models use standard scaling and drop a reference share (food) so the remaining coefficients are identified; tree models are unscaled. Robust scaling was dropped because zero-inflated shares have an interquartile range near zero, which inflated entertainment shares to values above 150.

The feature checks behave as before: expense-to-income ratios transfer worst across income halves on a neutral target (`Has_Bank_Savings`) even after winsorising (mean ROC-AUC 0.603 against 0.628 for raw rupees and 0.618 for shares); participation indicators lift ROC-AUC from 0.917 to 0.919;

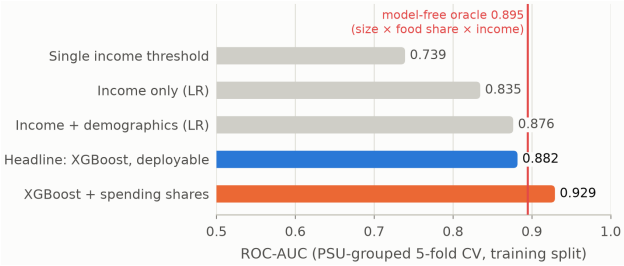



Fig. 1. Grouped-CV ROC-AUC on the training split as features and capacity are added. The red line is the score from household size, food share and income, with no fitted model: it beats every model that does not see spending shares, and the full model's extra 0.047 AUC over the headline runs largely through that route.

TABLE I
FIVE-FOLD PSU-GROUPED CROSS-VALIDATION ON THE TRAINING SPLIT (33,219 HOUSEHOLDS). XGBOOST (TUNED) USES RESULTS/MODEL_FINAL.JSON; "FULL" ADDS THE SPENDING SHARES AND IS DIAGNOSTIC ONLY. UNTUNED XGBOOST, LINEAR SVM AND THE DECISION TREE (MACRO-F1 0.774, 0.774, 0.751) ARE IN RESULTS/MODEL_COMPARISON.CSV.

| | Deployable (headline) | | | | Full |
|---|---|---|---|---|---|
| Model | F1 | AUC | PR-AUC | ECE | AUC |
| **XGBoost (tuned)** | **0.782** | **0.882** | 0.942 | 0.011 | 0.929 |
| HistGradientBoosting | 0.778 | 0.880 | 0.941 | 0.012 | 0.928 |
| Logistic Regression | 0.773 | 0.876 | 0.938 | 0.103 | 0.920 |
| Random Forest | 0.777 | 0.870 | 0.934 | 0.082 | 0.915 |
| Single income threshold | 0.742 | 0.739 | 0.808 | 0.003 | 0.739 |
| Majority class | 0.405 | 0.500 | 0.681 | 0.319 | 0.500 |

and a centred log-ratio transform of the shares loses (0.911 against 0.920). A missing debt value is informative (goal attainment 0.367 against 0.315), which the indicator now captures.

Macro-F1 remains the operational ranking metric, but every model is also scored on threshold-free metrics: ROC-AUC, PR-AUC for the at-risk class (positive $= 1 -$ `Goal_Met`, prevalence 0.681), Brier score, log-loss and expected calibration error (ECE).

## IV. RESULTS

### A. What each ingredient is worth

The majority classifier reaches 0.681 accuracy and 0.405 macro-F1. A depth-1 tree on income, re-learned inside every training fold (threshold Rs. 110,377–126,545, mean Rs. 122,249 a year), reaches 0.778 accuracy and 0.742 macro-F1. Logistic regression on income alone reaches ROC-AUC 0.835, and 0.876 with demographics and debt. Adding the food share to income lifts it to 0.875, and to 0.894 once household size is also present, which is the reconstruction route at work (Fig. 1).

### B. Model comparison

On the deployable set the tuned XGBoost model (`max_depth`=4, `learning_rate`=0.08, `n_estimators`=200, `subsample`=0.7, `min_child_weight`=1) leads at macro-F1 0.782 and ROC-AUC 0.882 (Table I); histogram gradient boosting is within one fold standard deviation. Tuning was worth 0.007 macro-F1, and depth was the parameter that mattered. PR-AUC is reported for the at-risk class against its 0.681 base rate. The linear models rank close on ROC-AUC but are poorly calibrated (ECE 0.10) because of balanced class weights.

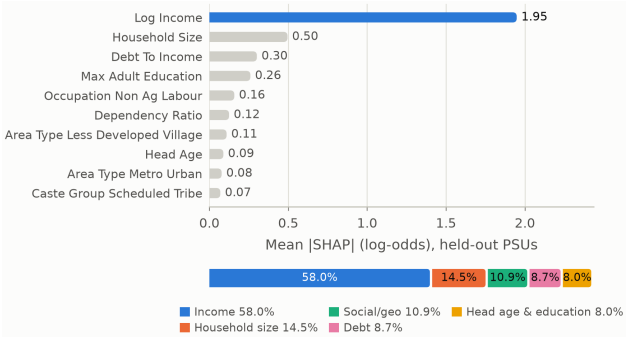



Fig. 2. SHAP for the headline model on held-out PSUs: the ten largest mean absolute values, and attribution by feature family, computed by summing SHAP values within each family before averaging.

On the held-out PSUs, touched once, the headline model scores ROC-AUC 0.886, macro-F1 0.781, at-risk PR-AUC 0.944, Brier 0.127 and ECE 0.009; the full model scores 0.932 and 0.836. A household-level random split gives almost the same full-model score (0.931), so the gap between the two models comes from the feature set, not from the validation scheme.

For deployment the at-risk threshold is tuned on out-of-fold training scores, not accepted at 0.5. The F1-optimal threshold, applied once to the held-out set, gives at-risk precision 0.826 and recall 0.913. Out of fold, raising at-risk recall from 80% to 95% costs 10 points of precision (0.893 to 0.789).

### C. What the model uses

We attribute with `shap.TreeExplainer`, verified to reproduce the raw margin to $6.5 \times 10^{-6}$. Because SHAP values are additive, a family's effect is the sum of its columns' values; we average its absolute value, which avoids double counting columns that push in opposite directions. Income carries 58.0% of grouped attribution in the headline model, the household-size family 14.5%, the social and geographic categoricals 10.9%, debt 8.7% and head age with education 8.0% (Fig. 2). Grouped permutation importance agrees on the order and is more extreme: permuting income costs 0.385 ROC-AUC, 82% of the total drop. Interactions are 31.6% of total attribution over the whole held-out set, led by size with income.

In the full model the shares family carries 19.1% of grouped SHAP (16.0% of permutation importance), and the strongest interaction is food share with income. Read naively, this says that a food-dominated budget signals adequacy once income is controlled. The mechanism is the reconstruction route: at a

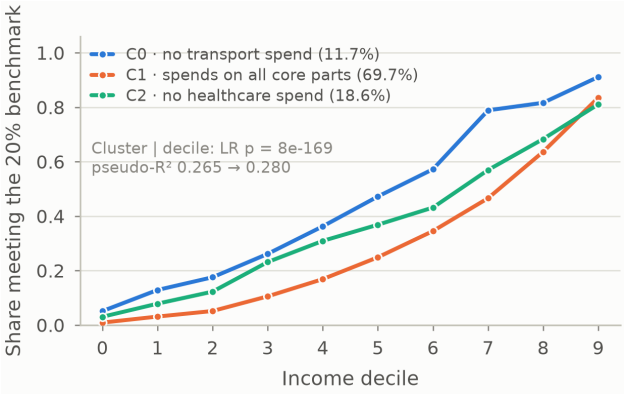



Fig. 3. Goal attainment by spending cluster within income decile (all ten). Cluster membership adds explanatory power beyond income (likelihood-ratio test), but the clusters are defined by which categories are absent.

given size and income, a higher food share implies a smaller total budget. The food-share SHAP values rise as the implied budget-to-income ratio falls (Spearman 0.27).

The shares are also not identified in a linear model. Refitting on five disjoint subsamples, three of ten share coefficients flip sign with all eleven shares included and one with food dropped as reference. With standard scaling, dropping the reference changes no coefficient's spread by more than a factor of 1.4 in either direction. Robust scaling is avoided because a zero-inflated share such as rent has an interquartile range of zero.

### D. Spending clusters

We clustered on one representation only, five isometric log-ratio coordinates of the six least zero-inflated shares, and chose $k$ within it. Silhouette (0.412), Davies–Bouldin (0.98) and Calinski–Harabasz all select $k = 3$, and the partition is stable under bootstrap resampling (median adjusted Rand index 0.998). It is not, however, a set of behavioural personas. The clusters are households with no transport spend (11.7%), with no healthcare spend (18.6%), and the rest (69.7%); their adjusted Rand index with the raw zero pattern is 0.858. Varying the zero-replacement constant from 0.1 to 1 times the smallest positive share keeps the partition (index 0.93–1.00 against the baseline) but moves the silhouette that justifies it from 0.39 to 0.48.

The clusters are close to independent of income (normalised mutual information with income decile 0.012; Kruskal–Wallis $\epsilon^2 = 0.034$). Unconditionally their goal attainment differs by 9.2 points (Cramér's $V = 0.076$). Conditioning on income decile, cluster membership is significant (likelihood ratio 774 on 2 df; pseudo-$R^2$ 0.265 to 0.280), with an income-adjusted odds ratio of 0.34 for the spend-on-everything cluster against the no-transport cluster, and the effect varies by decile (interaction $p = 10^{-19}$; Fig. 3). The mechanism is the reconstruction route again: a household with an empty category spends less in total at a given income.

TABLE II
TARGETING AT A 25% BUDGET (CEILING 36.7%). MODELS ARE FITTED UNWEIGHTED; THE "WEIGHTED" COLUMNS APPLY SURVEY WEIGHTS TO THE SAME SCORES AND ESTIMATE THE POPULATION.

| | Sample | | | Weighted | |
|---|---|---|---|---|---|
| Strategy | Recall | Precision | Ceiling | Recall | Precision |
| **Headline model** | **0.362** | **0.985** | 0.985 | 0.354 | 0.985 |
| Poorest first | 0.350 | 0.954 | 0.954 | 0.342 | 0.951 |
| Random contact | 0.249 | 0.678 | 0.678 | 0.248 | 0.691 |

### E. What the model is worth

At a 25% contact budget no strategy can reach more than 36.7% of at-risk households, because they are 68.1% of the population. Table II separates three quantities: *recall* (the share of all at-risk households reached), *precision* (the share of contacts that are at risk) and *ceiling capture* (recall over the ceiling $\min(1, b/\text{prevalence})$). Below the at-risk prevalence ceiling capture equals precision, because both are hits per contact, so they are one number under two names and separate only above a 68% budget. The model's recall gain over a poorest-first rule is 1.1 points at 25% (PSU-bootstrap 95% CI 1.0–1.3) and 0.3, 2.8 and 2.7 at 10%, 50% and 65% (Fig. 4(a)). A deployed threshold is a different operating point: out of fold it contacts 75.3% of households at recall 0.914 and precision 0.826 (held out: 0.913, 0.826).

Weighting matters for prevalence (at-risk 69.5% weighted against 68.1%) but not for the model: refitting with survey weights moves out-of-fold ROC-AUC from 0.8829 to 0.8811 and weighted recall at 25% from 0.3540 to 0.3536, so we keep the unweighted fit and use weights only for population figures. Calibration is good (out-of-fold ECE 0.005). Accuracy is lowest where the decision is contested: 0.98 in the poorest decile, 0.83 in the richest and 0.66 in decile 7 (Fig. 4(b)).

Peer benchmarks cannot locate recoverable spending. In rupees at-risk households out-spend same-income on-track peers in every category, which is mechanical; in budget shares they give more to healthcare ($+4.4$ points) and education ($+2.8$) and less to food ($-9.0$). Whether they could close their gap by spending like peers is 92.8% or 22.3% depending on the benchmark, so IHDS cannot split structural from behavioural shortfall (notebook 07). Ranking by a quantile model's predicted rupee gap reaches 45.5% of the total gap at 25% against 37.0% for the classifier and 28.7% for poorest-first.

### F. Outreach under assumed response

IHDS records no outreach, so response to a nudge cannot be estimated from it; we ask instead how much the targeting decision depends on that unknown. A contacted household responds with probability $r$; a responder raises its savings rate by $\delta = 5$ points of income; it is moved to on-track only if it was within $\delta$ of the benchmark (4.6% of at-risk households); each contact costs Rs. 150. All are assumptions and are varied in the repository: $r$ uniform at 0.20, rising with income rank from 0.05 to 0.35 (access constraints [7]), or 0.10 against 0.30

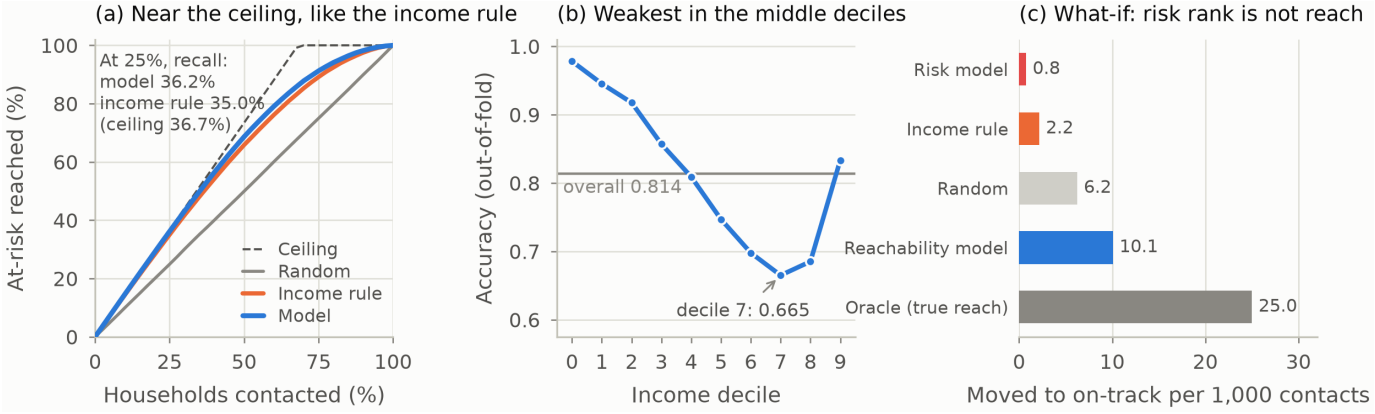



Fig. 4. Three views of deployment value, all out-of-fold under PSU-grouped CV. (a) At-risk households reached (recall) against contact budget, with the ceiling $\min(1, b/0.681)$. (b) Accuracy of the headline model by income decile. (c) Outreach what-if at a 25% budget, uniform response 0.20, a 5-point nudge: households moved to on-track per 1,000 contacts. *Response is assumed, not identified.* Income-gradient and debt-burdened responses give 0.4/1.1/7.2/11.8/28.4 and 0.6/2.5/6.6/10.7/27.4 in the same order.

for indebted households; $\delta$ of 2.5 and 10 points. We compare the risk model, poorest-first, random contact and a *reachability* model: the headline XGBoost fitted to the benchmark and to the benchmark relaxed by $\delta$, whose difference in $P(\text{on track})$ estimates the chance of lying in the band a nudge could cross.

Per 1,000 contacts at a 25% budget and uniform response (Fig. 4(c)) the risk model moves 0.8 households, poorest-first 2.2, random contact 6.2, the reachability model 10.1 and an oracle that knows each savings rate 25.0. A moved household costs Rs. 199,615 under the risk ranking, Rs. 24,028 at random and Rs. 14,885 under reachability; a programme breaks even only if a moved household is worth more than that to it. The reachability edge over the risk model is 9.3 per 1,000 (PSU-bootstrap 95% CI 8.4–10.1), 11.3 (10.3–12.2) if response rises with income and 10.1 (8.9–11.0) if debt lowers it, and the ordering holds at 2.5 and 10 points and with survey weights. The most at-risk are the deepest dissavers, out of reach of a small nudge, which is why a risk ranking can do worse than random here. This is a sensitivity analysis, not an estimate of benefit: the survey identifies neither $r$ nor $\delta$, a different objective such as rupees of gap closed would reorder the strategies, and an uplift estimate needs campaign data.

### G. External validity

No public survey records income and consumption for the same households after 2011–12, so the classifier cannot be tested out of time. Three checks stand in. *Mechanism:* HCES 2022–23 [17] (261,746 households, no income, so the label cannot be rebuilt) shows the ingredient of the leak. Food spend rises with household size with elasticity 0.57 against 0.63 in IHDS-II, residual sd 0.47 against 0.49, and mean food share 0.465 against 0.466. A simulation of the oracle over elasticities and residual sds places both surveys at an implied AUC of 0.87 (empirical IHDS-II 0.895), so the leak is a property of how tightly food tracks size, not of this file. Nominal per-capita spending grew 2.4–2.5 times, so any income threshold is a 2011–12 rupee figure. *Transport:* holding out whole states in five folds gives ROC-AUC 0.871–0.890 (grouped CV 0.882), and the model beats poorest-first at 25% in every fold (+0.7 to +1.6 recall points). *Calibration:* mean predicted minus observed on-track rate ranges from $-7.1$ to $+3.9$ points across state folds, and is $+9.1$ rural-to-urban and $-4.8$ urban-to-rural. Rankings travel; probabilities need recalibration.

## V. Discussion

A model trained on spending shares scores 0.93 ROC-AUC, but a three-column formula with no fitted parameters beyond a median scores 0.895, and removing the shares costs only 0.047. The model we recommend uses what a team would know at onboarding: income, household composition, debt and social group. It reaches 0.886 on held-out sampling units and improves on a poorest-first rule by about one percentage point of at-risk recall at a 25% budget, with an interval that excludes zero. That gain is small, consistent with Brown et al.'s experience of proxy means tests [2].

Four general lessons follow. First, a feature can pass every univariate leakage check and still reconstruct the label jointly with others; composition shares do so whenever one category is close to proportional to a measured covariate. Spending shares belong in a deployed model only if they are observed before the prediction window, which is how the scoring service in the repository enforces them. Second, capture should be reported against its ceiling and against the obvious rule; with a majority at-risk class, lift over random is uninformative. Third, peer-benchmark findings about "recoverable" spending are only as good as the benchmark, and here two reasonable constructions disagree by a factor of four. Fourth, ranking by risk and ranking by response are different decisions: under every assumed response the most at-risk households are the least movable, so a model that captures them well can still be a poor guide to a nudge.

Income under-reporting remains the largest limitation. Consumption exceeds income for 55.9% of households (57.7% weighted), 22.0% report spending more than twice their income, and all of the latter are at risk, 32.3% of that group.

This is a documented property of Indian surveys [8]; relative rankings hold and absolute prevalence claims do not. The data is from 2011–12 and the leakage mechanism, but not the absolute probabilities, survives in 2022–23; the unit is a household; model fitting is unweighted while population figures use the survey weights, which changes nothing material; debt is a stock with no repayment flow; and the 20% benchmark is a convention. IHDS records no intervention, so the outreach results are assumption-driven and estimate no one's response; an uplift model needs campaign data.

## VI. Reproducibility

Code, walkthroughs and result artifacts (MIT licence) are at https://github.com/saint2706/savings-goal-classifier. ICPSR's terms prohibit redistributing the IHDS-II microdata, so three routes are provided. (i) `uv run sgc demo` runs build, features, grouped CV, targeting and the outreach what-if on a synthetic survey with the real schema in about 15 seconds, with no registration; CI runs it on every push, and its numbers are not results. (ii) `uv run sgc reproduce --tsv <path>` rebuilds the data from the free ICPSR download, runs the ten notebooks and the figure script (about 25 minutes on four cores), and writes a manifest of package versions, git commit, seed and the input file's SHA-256 (ours begins 5ee026cc), so a reader can confirm they hold the same file. (iii) Every table and figure is regenerated from committed aggregate files with no microdata, and the README maps each claim to its script and artifact. The environment is pinned by `uv.lock` (Python 3.11–3.13), every seed is 42 and every split is grouped by PSU. The build is validated against a Pandera schema, and the package is tested at 99% coverage with ruff and mypy in CI. HCES files are read from the public dump in [17], with instructions in `dataset/README.md`; our loader reproduces the published monthly per-capita expenditure (Rs. 3,773 rural, Rs. 6,459 urban).

### A. AI Use Declaration

Claude Code (Anthropic), an AI coding assistant, was used to write the analysis package, notebooks, tests and figure script, and to draft this report's text from the resulting artifacts. We verified the numbers against the committed result files and revised the text. The analysis design and interpretation are our own, as recorded in the walkthroughs. We make no claim about the proportion of AI-generated text.

## VII. Conclusion

We built a savings-adequacy classifier for 41,518 Indian households and found that spending composition, the representation usually chosen to avoid leakage, leaks through household size. The model we recommend uses income, demographics and debt, reaches ROC-AUC 0.886 on held-out sampling units, and adds about a percentage point of at-risk recall over contacting the poorest first. It belongs inside an income-ranked campaign, not in front of one, and the survey cannot say whether the households it ranks first are the ones a nudge would move: under assumed responses they are not.

The broader result is methodological. A normative savings threshold turns most spending-based features into restatements of the label, singly or in combination, and conclusions drawn from them need a multivariate leakage check, grouped validation and uncertainty intervals before they can be trusted. Here those checks rule out a reversed Engel effect, a single miscellaneous-spending lever, and any firm split between structural and behavioural shortfall, and a simulation and a second survey show when the leak should be expected.


## Acknowledgment

We thank our course instructor for feedback on the framing of this project, and NCAER and ICPSR for collecting and distributing the IHDS-II microdata.


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